

JEE Mains 2019 Chapter wise Question Bank

Units and Dimensions - Questions

Q1

Expression for time in terms of G (universal gravitational constant), h (Planck constant) and c (speed of light) is proportional to:

- (1) $\sqrt{\frac{hc^5}{G}}$ (2) $\sqrt{\frac{c^3}{Gh}}$
 (3) $\sqrt{\frac{Gh}{c^5}}$ (4) $\sqrt{\frac{Gh}{c^3}}$

9 Jan Evening

Q2

The density of a material in SI unit is 128 kg m^{-3} . In certain units in which the unit of length is 25 cm and the unit of mass is 50 g, the numerical value of density of the material is:

- (1) 40 (2) 16 (3) 640 (4) 410

10 Jan Morning

Q3

The force of interaction between two atoms is given by

$$F = \alpha \beta \exp\left(-\frac{x^2}{\alpha k T}\right); \text{ where } x \text{ is the distance, } k \text{ is the}$$

Boltzmann constant and T is temperature and α and β are two constants. The dimension of β is:

- (1) $M^0 L^2 T^{-4}$ (2) $M^2 L T^{-4}$
 (3) $M L T^{-2}$ (4) $M^2 L^2 T^{-2}$

11 Jan Morning

Q4

If speed (V), acceleration (A) and force (F) are considered as fundamental units, the dimension of Young's modulus will be :

- (1) $V^{-2} A^2 F^{-2}$ (2) $V^{-2} A^2 F^2$
 (3) $V^{-4} A^{-2} F$ (4) $V^{-4} A^2 F$

11 Jan Evening

Q5

Let l , r , c and v represent inductance, resistance, capacitance and voltage, respectively. The dimension of $\frac{\ell}{rcv}$ in SI units

will be :

- (1) $[L A^{-2}]$ (2) $[A^{-1}]$
 (3) $[L T A]$ (4) $[L T^2]$

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Q6

In SI unit the dimensions of $\sqrt{\frac{\epsilon_0}{\mu_0}}$ is :

- (1) $A^{-1} T M L^3$ (2) $A T^2 M^{-1} L^{-1}$
 (3) $A T^{-3} M L^{3/2}$ (4) $A^2 T^3 M^{-1} L^{-2}$

8 April Morning

Q7

If Surface tension (S), Moment of Inertia (I) and Planck's constant (h), were to be taken as the fundamental units, then dimensional formula for linear momentum would be :

- (1) $S^{1/2} I^{3/2} h^{-1}$ (2) $S^{1/2} I^{1/2} h^{-1}$
 (3) $S^{1/2} I^{1/2} h^0$ (4) $S^{3/2} I^{1/2} h^0$

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Q8

In the formula $X = 5YZ^2$, X and Z have dimensions of capacitance and magnetic field, respectively. What are the dimensions of Y in SI units ?

- (1) $[M^{-3} L^{-2} T^8 A^4]$ (2) $[M^{-1} L^{-2} T^4 A^2]$
(3) $[M^{-2} L^0 T^{-4} A^{-2}]$ (4) $[M^{-2} L^{-2} T^6 A^3]$

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Q9

Which of the following combinations has the dimension of electrical resistance (ϵ_0 is the permittivity of vacuum and μ_0 is the permeability of vacuum) ?

- (1) $\sqrt{\frac{\mu_0}{\epsilon_0}}$ (2) $\frac{\mu_0}{\epsilon_0}$ (3) $\sqrt{\frac{\epsilon_0}{\mu_0}}$ (4) $\frac{\epsilon_0}{\mu_0}$

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Units and Dimensions - Answers

Q1

(3) Let $t \propto G^x h^y C^z$ Dimensions of $G = [M^{-1}L^3T^{-2}]$, $h = [ML^2T^{-1}]$ and $C = [LT^{-1}]$ $[T] = [M^{-1}L^3T^{-2}]^x [ML^2T^{-1}]^y [LT^{-1}]^z$ $[M^0L^0T^1] = [M^{-x+y}L^{3x+2y+z}T^{-2x-y-z}]$

By comparing the powers of M, L, T both the sides

$$-x + y = 0 \Rightarrow x = y$$

$$3x + 2y + z = 0 \Rightarrow 5x + z = 0 \dots (i)$$

$$-2x - y - z = 1 \Rightarrow 3x + z = -1 \dots (ii)$$

Solving eqns. (i) and (ii),

$$x = y = \frac{1}{2}, z = -\frac{5}{2} \therefore t \propto \sqrt{\frac{Gh}{C^5}}$$

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Q2

(1) Density of material in SI unit,

$$= \frac{128\text{kg}}{\text{m}^3}$$

Density of material in new system

$$= \frac{128(50\text{g})(20)}{(25\text{cm})^3(4)^3}$$

$$= \frac{128}{64} \times (20) = 40 \text{ units}$$

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Q3

(2) Force of interaction between two atoms,

$$F = \alpha \beta e^{\left(\frac{-x^2}{\alpha kT}\right)}$$

Since exponential terms are dimensionless

$$\therefore \left[\frac{x^2}{\alpha kT}\right] = M^0L^0T^0$$

$$\Rightarrow \frac{L^2}{[\alpha]ML^2T^{-2}} = M^0L^0T^0$$

$$\Rightarrow [\alpha] = M^{-1}T^2$$

$$[F] = [\alpha][\beta]$$

$$MLT^{-2} = M^{-1}T^2[\beta]$$

$$\Rightarrow [\beta] = M^2LT^{-4}$$

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Q4

(4) Let $[Y] = [V]^a [F]^b [A]^c$

$$[ML^{-1}T^{-2}] = [LT^{-1}]^a [MLT^{-2}]^b [LT^{-2}]^c$$

$$[ML^{-1}T^{-2}] = [M^bL^{a+b+c}T^{-a-2b-2c}]$$

Comparing power both side of similar terms we get,

$$b = 1, a + b + c = -1, -a - 2b - 2c = -2$$

solving above equations we get:

$$a = -4, b = 1, c = 2$$

$$\text{so } [Y] = [V^{-4}FA^2] = [V^{-4}A^2F]$$

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Q5

(2) As we know,

$$\left[\frac{\ell}{r}\right] = [T] \text{ and } [cv] = [AT]$$

$$\therefore \left[\frac{\ell}{rcv}\right] = \left[\frac{T}{AT}\right] = [A^{-1}]$$

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Q6

$$\begin{aligned}
 (4) \quad \left[\sqrt{\frac{\epsilon_0}{\mu_0}} \right] &= \sqrt{\frac{\epsilon_0^2}{\mu_0 \epsilon_0}} = \left[\frac{\epsilon_0}{\sqrt{\mu_0 \epsilon_0}} \right] \\
 &= \epsilon_0 C [LT^{-1}] \times [\epsilon_0] \quad \left[\because \frac{1}{\sqrt{\mu_0 \epsilon_0}} = C \right] \\
 \therefore F &= \frac{q^2}{4\pi\epsilon_0 r^2} \\
 \Rightarrow [\epsilon_0] &= \frac{[AT]^2}{[MLT^{-2}] \times [L^2]} = [A^2 M^{-1} L^{-3} T^4] \\
 \therefore \left[\sqrt{\frac{\epsilon_0}{\mu_0}} \right] &= [LT^{-1}] \times [A^2 M^{-1} L^{-3} T^4] \\
 &= [M^{-1} L^{-2} T^3 A^2]
 \end{aligned}$$

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Q7

$$\begin{aligned}
 (3) \quad P &= K S^a I^b h^c \\
 \text{or } [MLT^{-1}] &= [MT^{-2}]^a [ML^2]^b [ML^2 T^{-1}]^c \\
 \text{or } a + b + c &= 1 \\
 2b + 2c &= 1 \\
 \text{and } -2a - c &= -1 \\
 \text{On solving above equations, we get} \\
 a &= 1/2, b = 1/2, c = 0 \\
 \text{So, } P &= S^{1/2} I^{1/2} h^0
 \end{aligned}$$

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Q8

$$(1) X = 5YZ^2$$

$$\Rightarrow Y \propto \frac{X}{Z^2} \quad \dots(i)$$

$$X = \text{Capacitance} = \frac{Q}{V} = \frac{Q^2}{W} = \frac{[A^2 T^2]}{[ML^2 T^{-2}]}$$

$$X = [M^{-1} L^{-2} T^4 A^2]$$

$$Z = B = \frac{F}{IL} \quad [\because F = ILB]$$

$$Z = [MT^{-2} A^{-1}]$$

$$Y = \frac{[M^{-1} L^{-2} T^4 A^2]}{[MT^{-2} A^{-1}]^2}$$

$$Y = [M^{-3} L^{-2} T^8 A^4] \quad (\text{Using (i)})$$

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Q9

$$(1) \sqrt{\frac{\mu_0}{\epsilon_0}} = \sqrt{\frac{\mu_0^2}{\epsilon_0 \mu_0}} = \mu_0 c \quad \left(\frac{1}{\sqrt{\mu_0 \epsilon_0}} = c \right)$$

$$\mu_0 c \rightarrow MLT^{-2} A^{-2} \times LT^{-1}$$

$$\rightarrow ML^2 T^{-3} A^{-2}$$

$$\rightarrow \text{dimensions of resistance}$$

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